Yeonwook Roh, Ph.D.

Assistant Professor at Korea University, Sejong Campus

Department of Digital Healthcare Engineering, College of Science and Technology,
Biological Adaptability Lab
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Education and Work Experience

Assistant Professor (Sep. 2026 – Present)
Korea University, Sejong Campus
College of Science and Technology
Department of Digital Healthcare Engineering
Biological Adaptability Lab

Postdoctoral Fellow (Aug. 2024 – Aug. 2026)
University of California, San Diego
Department of Mechanical and Aerospace Engineering
Advisor: Prof. Michael T. Tolley
Project: Soft Robot for Locomotion in Seabed Media

Postdoctoral Fellow (Sep. 2023 – Aug. 2024)
Ajou University, Suwon, Korea
Department of Mechanical Engineering
Advisor: Prof. Daeshik Kang
Project: Brain Probe for Deep Brain Monitoring

Ph. D in Mechanical Engineering (Mar. 2019 – Aug. 2023)
Ajou University, Suwon, Korea
Advisor: Prof. Seungyong Han
Co-Advisor: Prof. Daeshik Kang, Prof. Je-sung Koh
Project: Soft Robotics, Flexible Electronics, and Cardiovascular Monitoring Systems
Thesis: Vital Signal Sensing through a Milliscale Soft Gripper of the Variable Stiffness

BA in History and BS in Mechanical Engineering (Mar. 2015 – Feb. 2019)
Ajou University, Seoul, Korea

HONORS AND AWARDS

Wiley Top Viewed Article Recognition -Wiley VCH GmbH (2026)

MIT Technology Review Innovators Under 35 Semifinalist - Massachusetts Institute of Technology (MIT) (2025)

Selection for Best of Advanced Engineering Materials - Wiley VCH GmbH (2024)

University President Award - Ajou University (2024)

Talent Award of Korea (Minister's Award) - Korean Ministry of Education (2023)

29th Samsung Electronics Paper Award - Samsung Electronics (2023)

Best Poster Award - Korean Society of Mechanical Engineers (2023)

Best Paper Award - Ajou University (2022)

26th Samsung Electronics Paper Award - Samsung Electronics (2020)

RESEARCH GRANTS

Sejong Science Fellowship (Overseas) **-** National Research Foundation of Korea (Sep. 2024 – Aug. 2025) – 70,000,000 KRW

Post-Doctoral Domestic Training (Fostering the Next Generation of Researchers) - National Research Foundation of Korea (Sep. 2023 - Aug. 2024) – 60,000,000 KRW

Research Subsidies for Ph.D. Candidates (Fostering the Next Generation of Researchers) - National Research Foundation of Korea (May 2021 - May 2023) - 40,000,000 KRW

MAJOR ACHIEVEMENTS

(+: First author, *: Corresponding author)

1. **Roh, Y.+**, Kim, M.+, Won, S.M.+, Lim, D., Hong, I., Lee, S., Kim, T., Kim, C., Lee, D., Im, S. and Lee, G., Kim, D., Shin, D., Gong, D., Kim, B., Kim, S., Kim, K.H., Koo, B.K., Seo, S., Koh, J.S.*, Kang, D.*, and Han, S.*, "Vital signal sensing and manipulation of a microscale organ with a multifunctional soft gripper", ***Science Robotics*** 6, eabi6774 (2021), (IF=27.5, JCR%=1%).

2. **Roh, Y.+**, Lee, S.+, Won, S.M.+, Hwang, S., Gong, D., Kim C., Hong, I., Lim, D., H. Kim, M., Kim, B., Kim, Kim, T., Im, S., Shin, D., Kim, U., Choi, J., Koh, J.S.*, Kang, D.*, and Han, S.*, "Crumple-recoverable electronics based on plastic to elastic deformation transitions", ***Nature Electronics*** 7, 66-76 (2023), (IF=42.3, JCR%=0.1%).

3. Hong, I.+, **Roh, Y.+**, Cho, J.+, Lee, S., Kang, M., Choi, D., Gong, D., An, H., Lim, D., Shin, D., Park, J., Kim, C., Kim, T., Kim, M., Im, S., Lee, J., Lee, G., Kim, U., Ko, S., Koh, J. S.*, Kang, D.*, Han, S.*, "Deployable electronics with enhanced fatigue resistance

for crumpling and tension", ***Science Advances*** 11, eadr3654 (2025), (IF=12.5, JCR%=8.5%).

4. Gong, D.+, Kang, M.+, Hwang, S.+, Cho, J., Hong, I., **Roh, Y.***, Han, S.*, "Fiber-reinforced origami electronics with high rigidity and flexibility for display applications", ***npj Flexible Electronics*** 9, 108 (2025), (IF=15.5, JCR%=1.5%).

5. **Roh, Y.+**, Kim, H.+, Kim, A.+, Ji. K. +, Kang, M., Gong, D., Im, S., Park, J., Bae, Y., Park, J., Koh, J.S., Han, S., Lee, E.*, and Kang, D.*, "Transient shuttle for a widespread neural probe with minimal perturbation", ***npj Flexible Electronics*** 8, 40 (2024), (IF=15.5, JCR%=1.5%).

Journal Publications

Published: 19 papers, including 5 major achievements
As First & Corresponding Author: 10 papers

(+: First author, *: Corresponding author)

Journal Publications

1. Gong, D.+, Kang, M.+, Hwang, S.+, Cho, J., Hong, I., **Roh, Y.***, Han, S.*, "Fiber-reinforced origami electronics with high rigidity and flexibility for display applications", ***npj Flexible Electronics*** 9, 108 (2025), (IF=15.5, JCR%=1.5%).

2. Hong, I.+, Lim, D.+, Kim, D.+, Hong, M.+, Kang, S., Ju, K., Oh, T., Hwang, S., **Roh, Y.**, Gong, D., Kwon, G., Kim, T., Im, C., Kim, E., Lee, J., Kim, S., Kim, J., Kim., S., Shim, K., Lee, J., Seo, S., Koh, J. S.*, Han, S.*, and Kang, D.*, "Breathable, wearable skin analyzer for reliable long-term monitoring of skin barrier function and individual environmental health impacts", ***Nature Communications*** 16, 1 (2025), (IF=15.7, JCR%=7%).

3. Hong, I.+, **Roh, Y.+**, Cho, J.+, Lee, S., Kang, M., Choi, D., Gong, D., An, H., Lim, D., Shin, D., Park, J., Kim, C., Kim, T., Kim, M., Im, S., Lee, J., Lee, G., Kim, U., Ko, S., Koh, J. S.*, Kang, D.*, Han, S.*, "Deployable electronics with enhanced fatigue resistance for crumpling and tension", ***Science Advances*** 11, eadr3654 (2025), (IF=12.5, JCR%=8.5%).

4. Gong, D.+, **Roh, Y.+**, Lee, J.+, Hwang, S, Kim, C., Ji, K., Kwon, G., Back, I., Shin, D., Lim, D., Hong, I., Lee, D., Koh, J.S.*, Kang, D.*, and Han, S.*, "Shape-tunable electronic composite for stimulus-customizable detection via neutral plane shifting.", ***Materials Horizons*** 12, 1303-1313 (2025), (IF=10.7, JCR%=11.1%).

5. Kim, T.+, Hong, I.+, Im, S.+, Rho, S.+, Kim, M., **Roh, Y.**, Kim, C., Park, J., Lim, D., Lee, D., Lee, S., Lee, J., Back, I., Lee, J., Seo, S., Kim, U., Cho, J., Hong, R. M., Kang, S., Koh, J.S. *, Han, S. *, and Kang, D.*, "Fly-by-Feel: Wing strain-based flight control

of flapping-wing drones through reinforcement learning, ***Nature Machine Intelligence*** 6, 992-1005 (2024), (IF=23.9, JCR%=0.3%).

6. **Roh, Y.+**, Kim, H.+, Kim, A.+, Ji. K. +, Kang, M., Gong, D., Im, S., Park, J., Bae, Y., Park, J., Koh, J.S., Han, S., Lee, E.*, and Kang, D.*, "Transient shuttle for a widespread neural probe with minimal perturbation.", ***npj Flexible Electronics*** 8, 40 (2024), (IF=15.5, JCR%=1.5%).

7. **Roh, Y.+**, Lim, D.+, Kang, M.+, Cho, J., Han, S.* and, Ko, S.H.* "Rigidity-tunable materials for soft engineering systems.", ***Advanced Engineering Materials*** 26, 2400563 (2024), (IF:3.3, JCR% =49.2%).

8. **Roh, Y.+**, Lee, S.+, Won, S.M.+, Hwang, S., Gong, D., Kim C., Hong, I., Lim, D., H. Kim, M., Kim, B., Kim, Kim, T., Im, S., Shin, D., Kim, U., Choi, J., Koh, J.S.*, Kang, D.*, and Han, S.*, "Crumple-recoverable electronics based on plastic to elastic deformation transitions", ***Nature Electronics*** 7, 66-76 (2023), (IF=40.9, JCR%=0.1%).

9. Im, S.+, Kim, T.+, Min, C., Kang, S., **Roh, Y.,** Kim, C., Kim, M., Kim, H., Shim, K., Koh, J.S., Han, S., Lee, J., Kim, D*., Kang, D*., Seo, S.*, "Real-time counting of wheezing events from lung sounds using deep learning algorithms: Implications for disease prediction and early intervention", ***PLoS One*** 18, e0294447 (2023), (IF=2.6, JCR%=32.2%).

10. **Roh, Y.+**, Lee, Y. +, Lim, D.+, Gong, D., Hwang, S., Kang, M., Kim, D., J., Cho, Kwon, G., Kang, D., Han, S.* and, Ko, S.H.*, "Nature's blueprint in bioinspired materials for robotics", ***Advanced Functional Materials*** 34, 2306079 (2024), (IF=19, JCR%=4.5%).

11. Kim, T.+, Hong, I.+, Kim, M.+ Im, S.+, **Roh, Y.**, Kim, C., Lim, J., Kim, D., Park, J., Lee, S., Lim, D., Cho, J., Huh, S., Jo, S., Kim, C., Koh, J.S.*, Han, S.*, Kang, D.*, 2023, "Ultra-stable and tough bioinspired crack-based tactile sensor for small legged robots", ***npj Flexible Electronics*** 7, 22 (2023), (IF=15.5, JCR%=1.5%).

12. Kim, T.+, Hong, I.+, **Roh, Y.+**, Kim, D., Kim, S., Im, S., Kim, C., Jang, K., Kim, S., Kim, S., Kim, M., Park, J., Gong, D., Ahn, K., Lee, J., Lee, G., Lee, H., Kang, J., Hong, J.M., Lee, S., Seo, S., Koo, B.K*., Koh, J.S.*, Kang, D.*, and Han, S.*, "Spider-inspired tunable mechanosensor for biomedical applications", ***npj Flexible Electronics*** 7, 12 (2023), (IF=15.5, JCR%=1.5%).

13. Lim, D.+, Hong, I.+, Park, S.U., Chae, J.W., Lee, S., Baac, H.W., Shin, C., Lee, J., **Roh, Y.**, Im, C., Park, Y., Lee, G., Kim, U., Ko, S.H., Han, S.*, and Won, S.M.*, "Functional encapsulating structure for wireless and immediate monitoring of the fluid penetration", ***Advanced Functional Materials*** 32, 2201854 (2022), (IF:19, JCR%=4.5%).

14. **Roh, Y.+**, Kim, M.+, Won, S.M.+, Lim, D., Hong, I., Lee, S., Kim, T., Kim, C., Lee, D., Im, S. and Lee, G., Kim, D., Shin, D., Gong, D., Kim, B., Kim, S., Kim, K.H., Koo, B.K., Seo, S., Koh, J.S.*, Kang, D.*, and Han, S.*, "Vital signal sensing and manipulation of a microscale organ with a multifunctional soft gripper", ***Science Robotics*** 6, eabi6774 (2021), (IF=27.5, JCR%=1%).

15. Lee, G.+, Choi, Y.W.+, Lee, T.+, Lim, K.S., Shin, J., Kim, T., Kim, H.K., Koo, B.K., Kim, H.B., Lee, J.G., Ahn, K., Lee, E., Lee, M.S., Jeon, J., Yang, H.S., Won, P., Mo, S., Kim, N., Jeong, H.M., **Roh, Y.**, Han, S., Koh, J.S., Kim, S.M.*, Kang, D.*, and Choi, M.*, "Nature-inspired rollable electronics", ***NPG Asia Materials*** 11, 67 (2019), (IF:8.3, JCR%=17.5%).

16. Kim, M.+, Choi, H.+, Kim, T., Hong, I., **Roh, Y.**, Park, J., Seo, S., Han, S.*, Koh, J.S*. and Kang, D.*, "FEP encapsulated crack-based sensor for measurement in moisture-laden environment", ***Materials*** 12, 1516 (2019), (IF:3.2, JCR%=25.5%).

17. Hong, I.+, <u>**Roh, Y.**</u>+, Koh, J.S.+, Na, S., Kim, T., Lee, E., An, H., Kwon, J., Yeo, J., Hong, S., Lee, K.T., Kang, D.*, Ko, S.H.*, and Han, S.*, "Semipermanent copper nanowire network with an oxidation-proof encapsulation layer", ***Advanced Materials Technologies*** 4, 1800422 (2019), (IF:6.2, JCR%=26.4%).

18. Hong, I.+, Lee, S., Kim, D., Cho, H., **Roh, Y.**, An, H., Hong, S., Ko, S.H*. and Han, S.*, "Study on the oxidation of copper nanowire network electrodes for skin mountable," flexible, stretchable and wearable electronics applications", ***Nanotechnology*** 30, 074001 (2018), (IF:2.8, JCR%=46.3%).

Journal Papers in Review & Preparation as first author

1. Lim, D.+, <u>**Roh, Y.+**</u>, Lee, D.+, Lim, K.+, Hong, I.**,** Kim, M., Shin, D., Kim, S., Kim, E., Kang, P., Jeon, C., Son, H., Seo, S., Kim, H., Koo, B., Koh, J.S.*, Kang, D.*, and Han, S.*, "Early diagnosis of embolism with a perivascular sensor", *Completed but not submitted.*

2. <u>**Roh, Y.**+</u>, Chae, S.+, Gravish, N.*., and Tolley, M. T.***,** "Locomotion strategy for a digging robot in muddy environments", *In preparation.*

Patent

1. Han, S., Kang, D., Koh, J.S., **Roh, Y.,** Lee. S., "Variable stiffness electronic composites and panels fabricated therefrom for bendable electronic devices, Registered, KR 1028975030000, Registered, 2025.12.03.

2. Kang, D., Koh, J.S., Han, S., Kim, T., Hong, I., **Roh, Y.**, Kim, D., Lee, D., "Pulse and voice measuring device with adjustable sensitivity", KR 1026301100000, Registered, 2024. 01. 23.

3. Han, S., Kang, D., Koh, J.S., **Roh, Y.,** Lim, D., "Soft gripper sensor", KR 1025468110000, Registered, 2023.06.19.

4. Kang, D., Lee., E., **Roh, Y.**, Kim, H., Kim, E., Ji, K., "Nerve probe transport device 1", KR 1020240041325, Pending, 2024.03.26.

5. Kang, D., Lee., E., **Roh, Y.**, Kim, H., Kim, E., Ji, K., "Nerve probe transport device 2", KR 1020240041325, Pending, 2024.03.26.

Conferences Presentations

1. Seo, H., Acome, E., Kellaris, N., Michell, S., Rohrbaugh, N., **Roh, Y.,** Gravish, N., and Tolley, T., "Opportunities for Soft Electrohydraulic Actuation for Burrowing in Dry and Submerged Granular Media", IEEE RoboSoft, 2026.

2. **Roh, Y.**, and Han, S., "Curmple-recoverable electronics based on plastic to elastic deformation transitions", The Polymer Society of Korea, 2024

3. **Roh, Y.**, Kim. M., Lim, D., Hong, I., Koh, J.S., Kang, D., and Han, S. "Vital signal sensing through conformal, variable stiffness gripping for interlinkage with a micro-scale organ", Korea Robotics Society, 2023

4. **Roh, Y**., Lee, S., Hwang, S., Koh, J., Kang, D., and Han, S., "Wrinkle and Electrode Damage self-healing device for electronic and optic application", Korean Society for Precision Engineering Autumn Conference, 2022

5. **Roh, Y.**, Lim, D., Hong, I., Gong, D., Kim, C., Lee, D., Im, S., Kim, D., Kim, B., Choi, D., Hong, M., Park, J., Koh, J.S., Kang, D., and Han, S., "Vital signal sensing through conformal variable stiffness gripping for interlinkage with a micro-scale organ", Materials Research Society Spring Meeting & Exhibit, Honolulu, USA, 2022.

Research Experience

Biomedical Applications

- Implantable Robotics and Electronics to Monitor the Cardiovascular Systems
- Micro-Scale Brain Probe for Deep Brain Monitoring
- Wearable Sensor System for Monitoring Physical Activity Signals

Soft Robotics

- Development of the Milliscale Gripper Based on Variable Stiffness
- Development of the Deployable Gripper Based on Pneumatic System
- Integration of Sensors for Soft Robotics (temperature, pressure, proximity sensor)
- Optimization of Soft Robot Locomotion in Underwater Sand and Mud Environments

Display Materials and Shape-Reconfigurable Displays

- Wrinkle-Flattening of Foldable Displays after Folding
- Development of the Shape-Reconfigurable Display Based on Origami Structure

- Development of the Wearable Touch Panel

Semiconductor Process

- Physical Vapor Deposition (Sputter Deposition, Evaporation Deposition)
- Photolithography Process
- Etching Process
- MEMS (Micro-Electro-Mechanical Systems) Technology

Reference

Prof. Seungyong Han (Ph.D. Advisor)
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E-mail: syhan01@postech.ac.kr

Prof. Daeshik Kang (Ph.D. Co-Advisor)
Pohang University of Science and Technology
80, Jigok-ro, Nam-gu, Pohang-si, Gyeongsangbuk-do,
Republic of Korea
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Pohang University of Science and Technology
80, Jigok-ro, Nam-gu, Pohang-si, Gyeongsangbuk-do,
Republic of Korea
E-mail: jskoh@postech.ac.kr

Prof. Michael T. Tolley (Postdoctoral Advisor)
University of California, San Diego
Franklin Antonio Hall, room 3102, 9500 Gilman Drive #0433, La Jolla, CA 92093-0433,
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